

Enzyme Kinetics & Glucose Meter – Part I; ChE

Name(s): _____

Fill in all sections

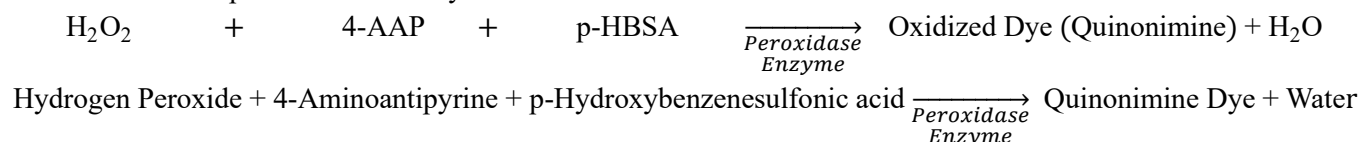
Note: This activity requires a cellphone with an RGB reader app. Color Assist for iPhone is recommended. Please download before beginning.

Student Learning Objectives

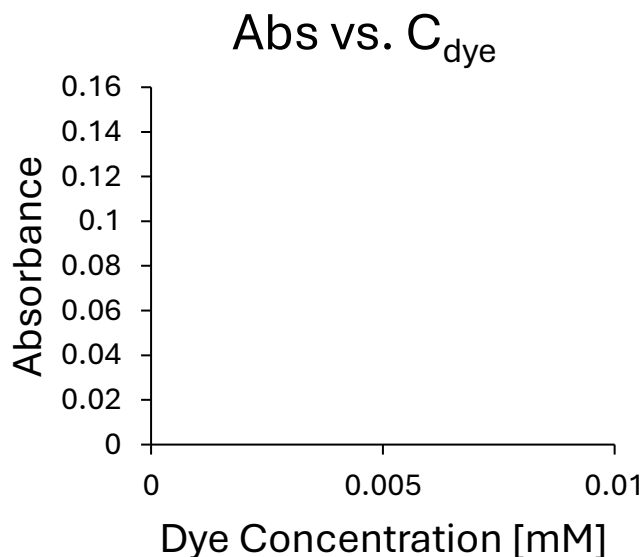
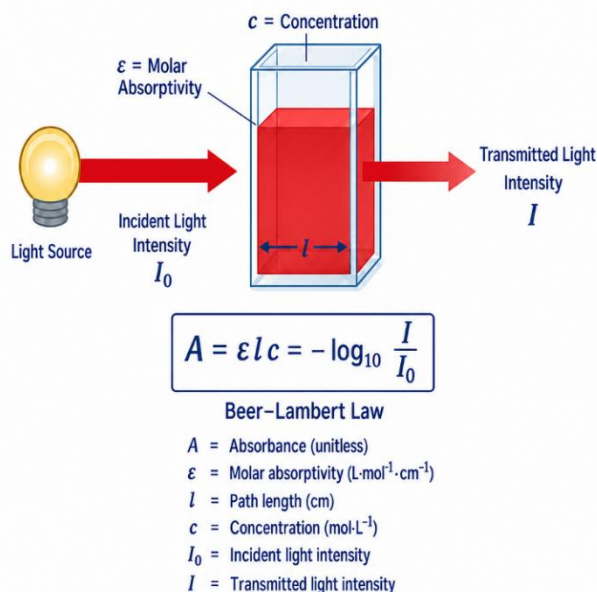
- 1) How to calibrate concentrations using Beer's Law
- 2) Understanding Michaelis-Menton enzyme kinetics for the Trinder reaction:
 - a. Understanding reaction stoichiometry and limiting reagent
 - b. Mass action effects on reaction rates
 - c. Enzyme catalyst turnover rate dictates maximum reaction rate
 - d. Use of absorbance vs. time graphs for understanding kinetics
 - e. Use of a kinetic read calibration curve
 - f. Linearizing the Michaelis-Menton equation for determining kinetic parameters

Before Experimenting with Your Physical or Virtual DLM: Calibration Curve & Beer's Law

We want to measure the product of this enzyme reaction:



From Beer's Law, the red colored quinonimine dye absorbs green light passing through a cuvette as shown below¹:



- 1) Based on the equation for Beer's law, graph the expected ABS vs. C_{Dye} result (roughly).

¹ Microsoft Copilot. May 2026. Beer-Lambert law diagram [AI-generated image].

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$$ABS = \epsilon \cdot L \cdot C = -\text{Log}(T) = -\log\left(\frac{I}{I_0}\right)$$

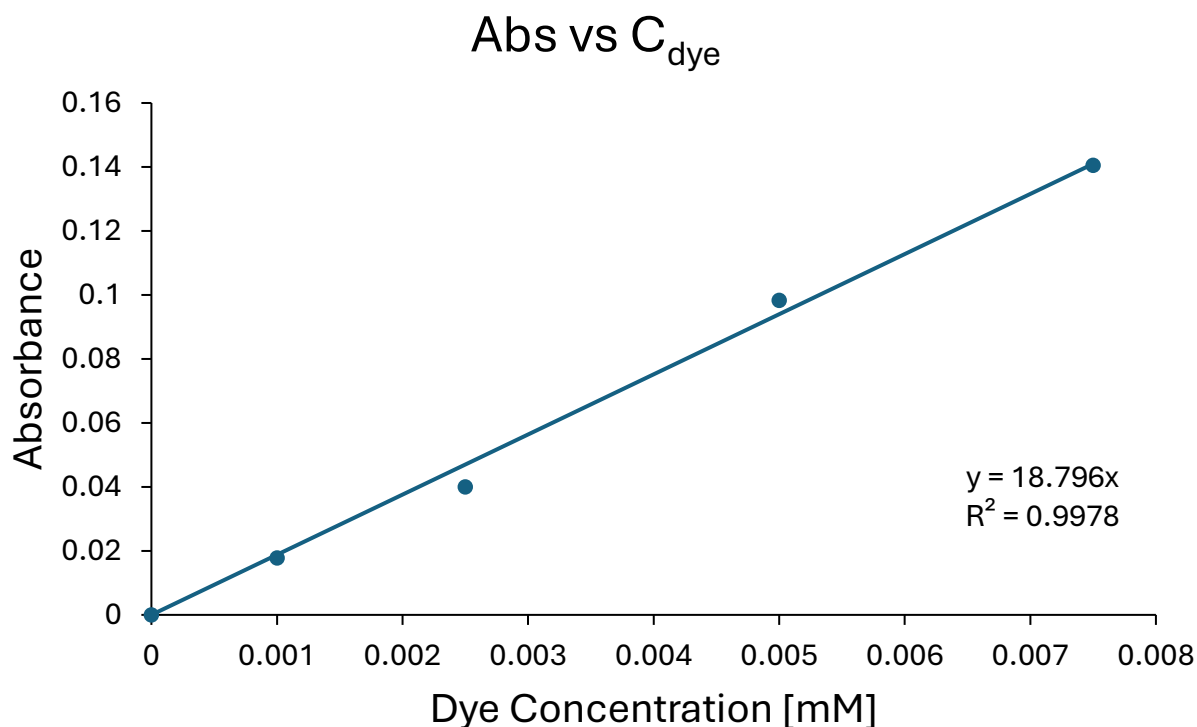
ϵ = molar absorptivity ($\sim 19.6 \text{ mM}^{-1}\text{cm}^{-1}$)

L = pathlength (0.635 cm)

C = concentration (known)

T = transmittance of green light through the sample

I = Measured intensity, I_0 = Blank Intensity



Reference using actual data from a spectrometer calibration curve.

Understand Michaelis-Menton Enzyme Kinetics

Michaelis-Menton rates are modeled as follows (see Kinetics class for derivation):

$$r_{Dye} = \frac{dC_{Dye}}{dt} = -\frac{dC_{H_2O_2}}{dt} = \frac{r_{max}C_{H_2O_2}}{K_M + C_{H_2O_2}}$$

Where:

$C_{H_2O_2}$ = Hydrogen peroxide reactant (substrate) concentration [mmol/liter, mM/L]

r_{max} = Maximum rate [mmol/s]

r_{Dye} = Reaction rate for dye formation [mmol/s]

K_M = Michaelis constant (0.032 mM)

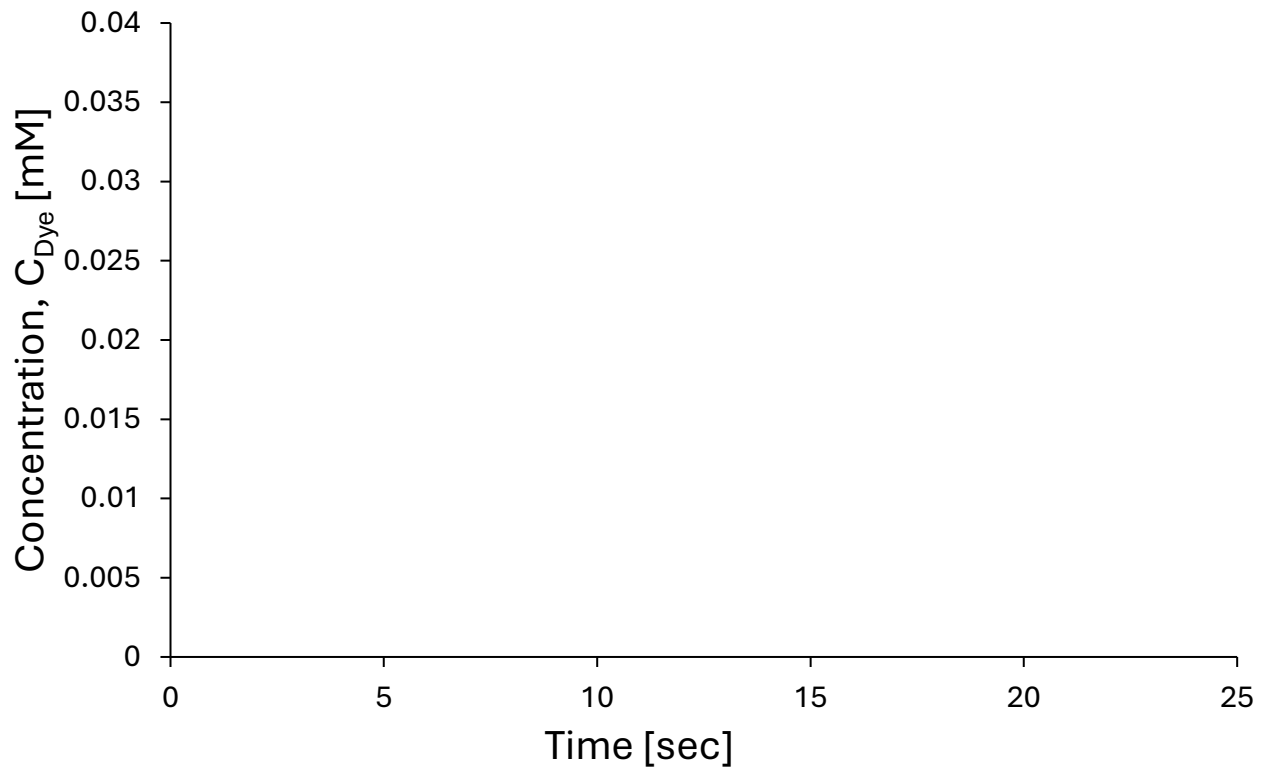
1) Given $K_M = 0.032$ and the 1:1 stoichiometry between H_2O_2 and the Dye, plot the expected curves for C_{Dye} vs. t when:

- $C_{H_2O_2} \ll K_M$, e.g., $C_{H_2O_2} = 0.001$ & 0.01 mM

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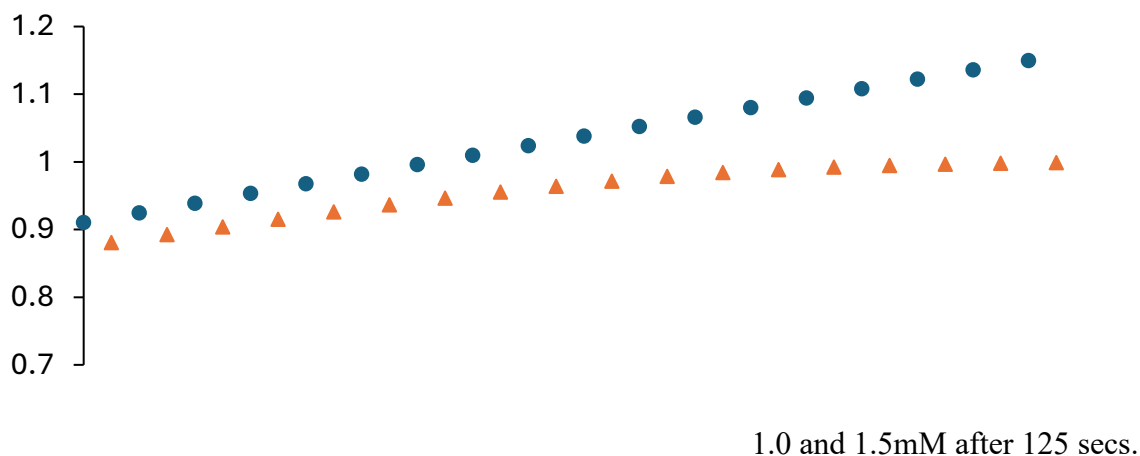
- $C_{H_2O_2} = K_M$, i.e., $C_{H_2O_2} = 0.032 \text{ mM}$
- $C_{H_2O_2} \gg K_M$, e.g., $C_{H_2O_2} = 1 \text{ \& } 3 \text{ mM}$
- 0.001, 0.01, 0.02, 0.032, 0.1, 1, & 3 mM

Dye Production

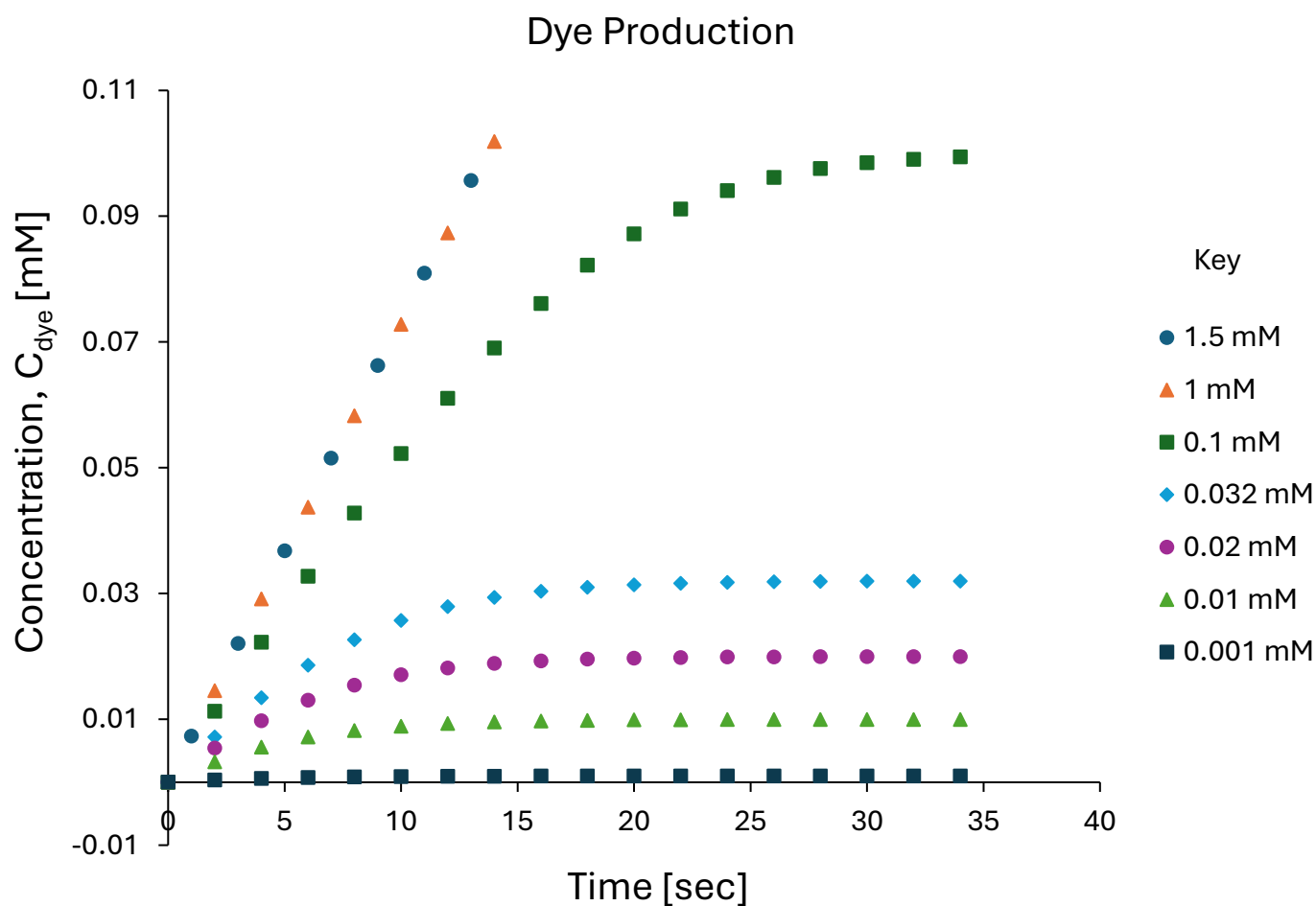


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Given lab data C_{Dye} [mM] vs. t [sec] below discuss answers for the questions below.



1.0 and 1.5mM after 125 secs.



Model of Michaelis-Menten rate for illustrative purposes.

2) Based on initial rates, $\left. \frac{dC_{Dye}}{dt} \right|_{t=0}$, and $r_{Dye} = \frac{dC_{Dye}}{dt} = \frac{r_{max}C_{H_2O_2}}{K_M + C_{H_2O_2}}$, answer the following:

a. At low $C_{H_2O_2} \ll K_M$, why do you see initial rates increasing? Hint: think about molecular collisions (mass action).

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- b. At high $C_{H_2O_2} \gg K_M$, why do you see the same initial rates? Hint: Think about enzyme processing rate and how an overabundance of reactant saturates all enzyme sites.

- c. Locate K_M on the graph based on the maximum rate. Remember its variables are in concentration (mM).

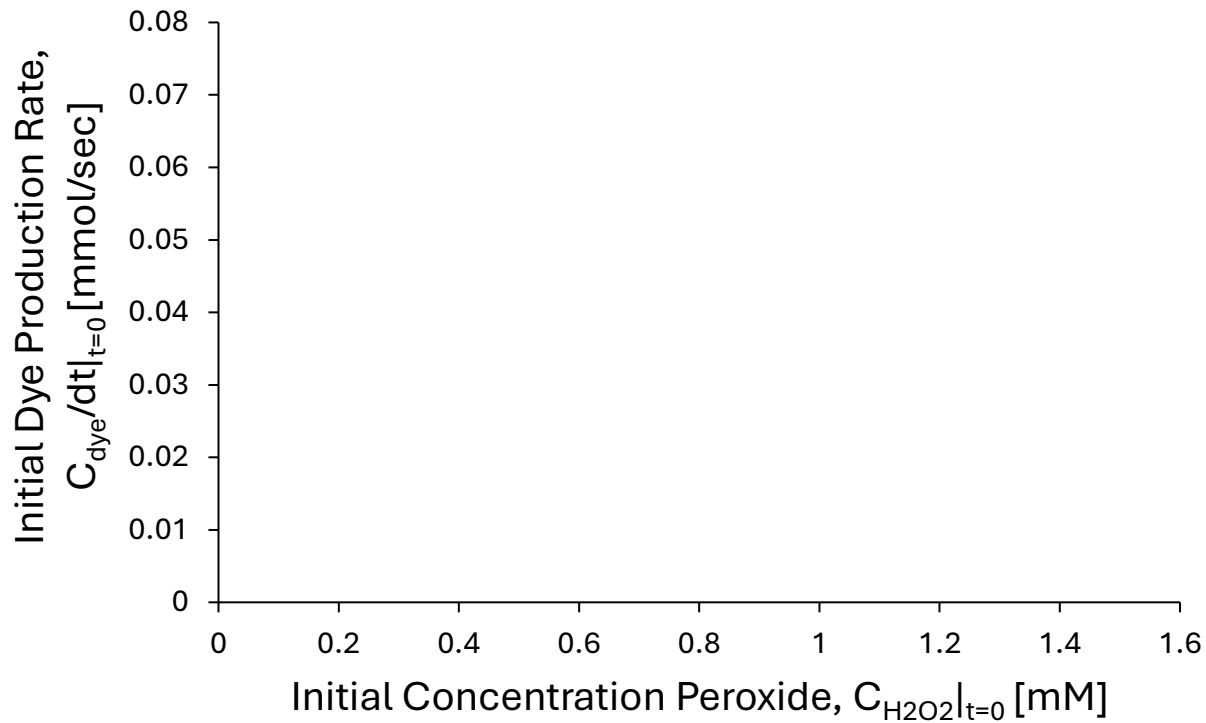
- d. Tell how you can get the initial rates from the data above, i.e., $\left. \frac{dC_{Dye}}{dt} \right|_{t=0} \left[\frac{mM}{sec} \right]$.

- e. Discuss why the plots level off with time. Hint: what happens to $C_{H_2O_2}$ as the reaction proceeds.

- 3) Now plot initial rates $\left. \frac{dC_{Dye}}{dt} \right|_{t=0}$ vs. $C_{H_2O_2} \big|_{t=0}$ from the table of data.

$C_{H_2O_2} \big _{t=0}$ [mM]	$\left. \frac{dC_{Dye}}{dt} \right _{t=0}$ [mmol/s]
0.001	0.000175
0.002	0.000343
0.003	0.000506
0.02	0.00258
0.03	0.00336
0.25	0.00665
0.75	0.00721
1.5	0.00736

Michealis Menten Example Graph



a) How do you get $r_{\max}$ i.e., $\left. \frac{dC_{\text{Dye}}}{dt} \right|_{\max}$ graphically? This is the enzyme turnover rate (= 0.0075 mmol/s).

b) Given, $r_{\text{Dye}} = \frac{dC_{\text{Dye}}}{dt} = \frac{r_{\max} C_{\text{H}_2\text{O}_2}}{K_M + C_{\text{H}_2\text{O}_2}}$, discuss how to get K_M quickly from the graph.

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Constructing Michaelis-Menten Kinetics Graph:

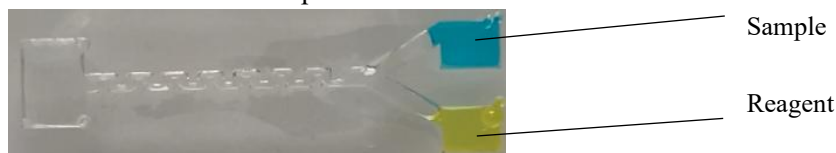
Step 1) Follow the set-up video for your kit. Wear gloves. Clean up spills.

Step 2) Add cellphone software (if you have not already): Color Assist for [iPhone](#). Color Picker for [Android](#).

Step 3) Use gloves and safety glasses for handling chemicals; record peroxide concentrations.

Step 4) With the module on its side load peroxide & enzyme solutions. Note your solutions are colorless; we used dyed water to show there are two separate liquids loaded into separate reservoirs.

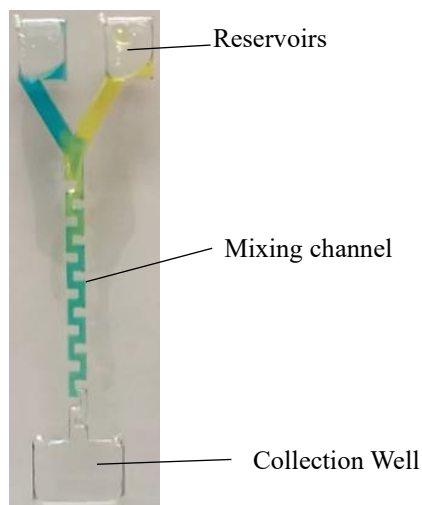
- To prevent premature reaction from leftover solutions, always load H_2O_2 solution on top, reagent solution on bottom for repeats.



- If loading is prevented (fluids failing to enter the reservoir, other reservoir contents gush out), place a gloved finger over opposite reservoir entrance to avoid leakage.

Step 5) Turn on the flashlight behind the cardboard box & insert your phone.

Step 6) Start a timer on a cell phone exactly when a teammate places module in an upright position with reservoirs at the top.



- If needed shake the module to allow the solutions to flow to the bottom.

Step 7) After 30 sec record the G value from the RGB readout below.

Step 8) Repeat steps 4-7 for a 2nd run & average results.

Step 9) Record the same measurements for a blank containing only enzyme solution in both chambers.

Step 10) Provide results to the professor who will post them on an overhead projector. Copy all results.

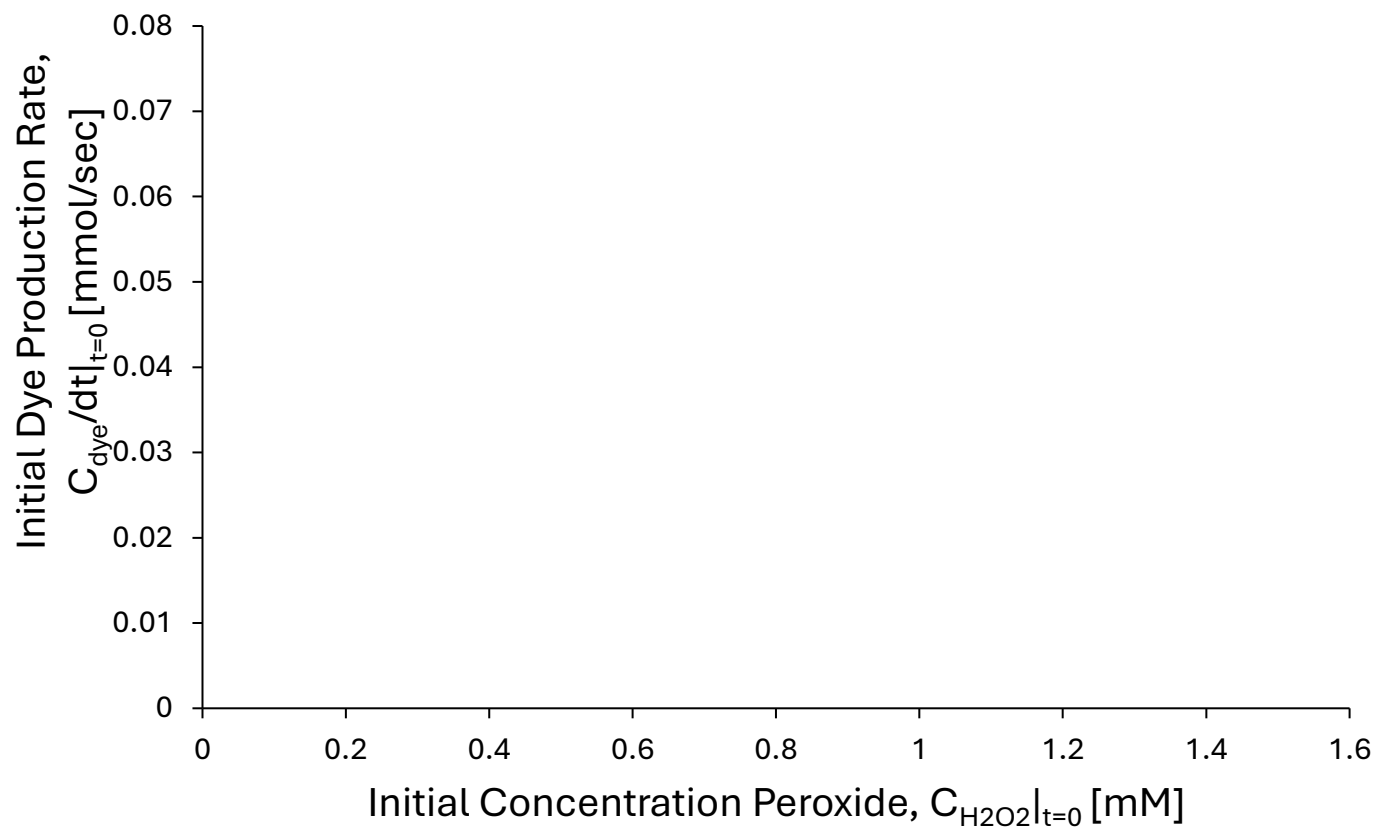
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$C_{H_2O_2}$ [mM]	t [sec]	Trial 1 G-Value (RGB)	Trial 2	Average	Blank
0.0050					
0.0075					
0.010					
0.015					
0.025					
0.032					
0.500					
0.750					
1.000					

Homework Problems:

Construct a graph of the initial rate of dye production $\frac{dC_{Dye}}{dt}\Big|_{t=0}$ vs. initial substrate concentration, $C_{H_2O_2}$, using the data collected in class. This will construct what’s called a Michaelis-Menten Plot.

Michealis Menten Lab Graph



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1. Identify the maximum rate, also known as the enzyme turnover rate.
 2. Locate the Michaelis constant on the graph. This will be the concentration at one half the maximum rate.
 3. Sketch out a Langmuir (Hanes-Woolf) plot using your data, plotting the initial concentration vs the concentration divided by the associated initial rate.
- Begin with the Michaelis-Menten Equation below;

- $$r = \frac{r_{max}C_{H_2O_2}}{K_M + C_{H_2O_2}}$$

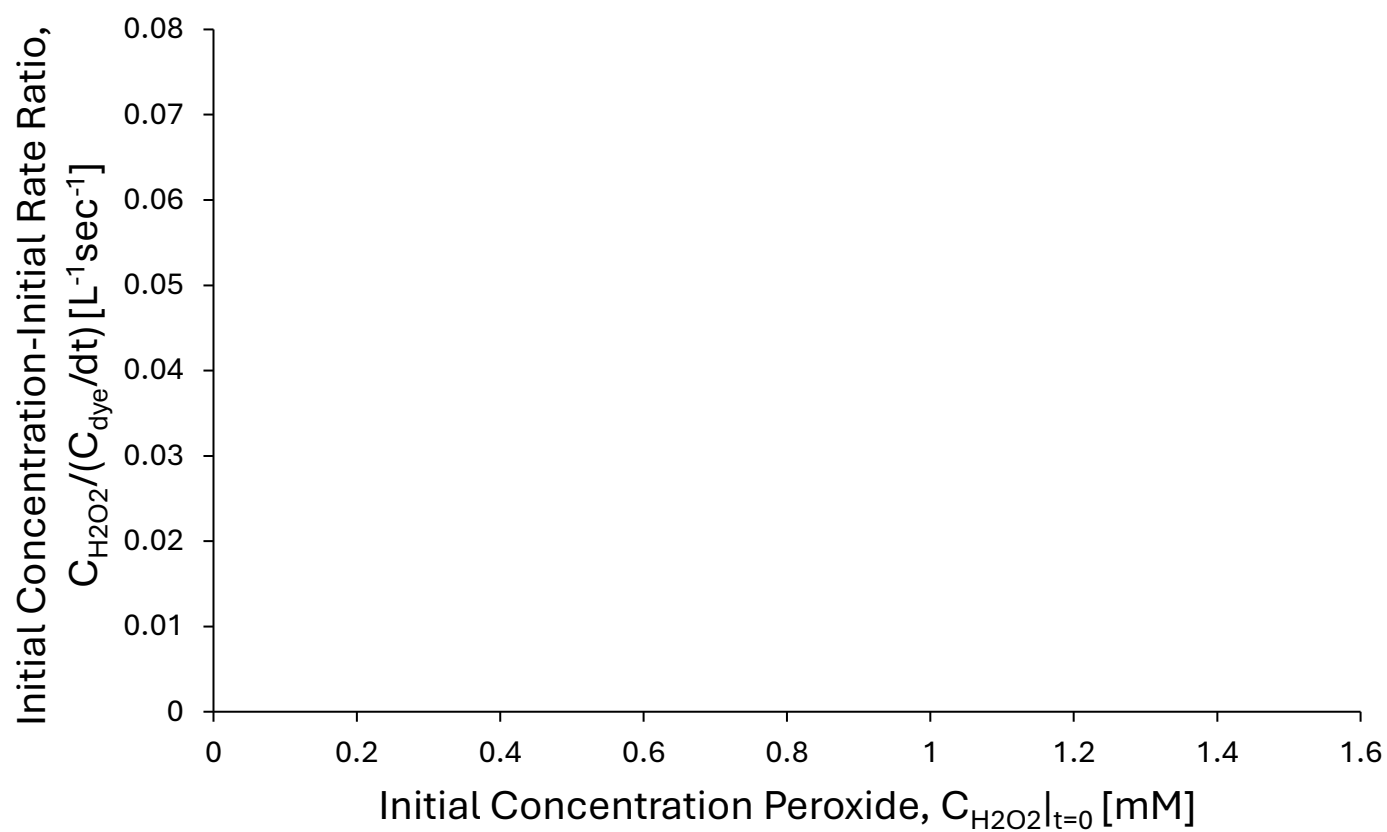
- Invert the Equation (show your work)

- Multiply by $C_{H_2O_2}$ (show your work)

- You Should Arrive at the following

- $$\frac{C_{H_2O_2}}{r} = \frac{K_M}{r_{max}} + \frac{C_{H_2O_2}}{r_{max}}$$

Langmuir Plot



4. With your new plot, identify the K_M and r_{max} .

Worksheet I: Enzyme Kinetics & Glucose Meter – Part I; ChE

Enzyme Kinetics & Glucose Meter – Part II; ChE

Fill in all sections – These are today's notes

Student Learning Objectives

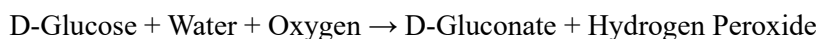
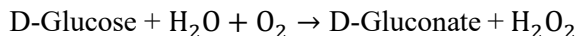
1. Understand stoichiometric principles in reaction design
 - a. Identify limiting reagents and mechanisms of delivery
2. Understand the utility of a probe reaction
3. Understand a kinetic vs. endpoint read
4. Examine complex reaction kinetics
 - a. Examine glucose oxidase reaction
 - b. Estimate rate constants via graph fitting
5. Understanding glucose reader kinetics & limiting reactant:
 - a. Mass action effects
 - b. Recognizing when a reactant is limiting
 - c. Use of absorbance vs. time graphs for understanding kinetics

Before Experimenting with Your Physical or Virtual DLM:

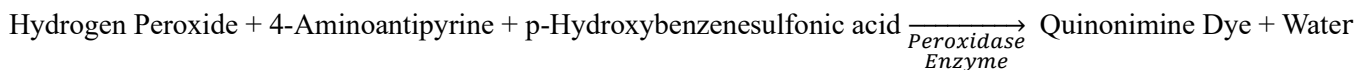
Please ensure that a member of your group has the appropriate software ready for the experiment downloaded onto their phone. If you have not done so already, do it now. For Apple iPhone, this would be the Color Assist app. For Android, Color Picker.

The Trinder Reaction and Stoichiometry

The Trinder reaction is a process that can roughly be broken into two phases. During the first phase, glucose reacts to form D-Gluconate and hydrogen peroxide using an enzyme called glucose oxidase:



You should be familiar with the second phase from Part 1. During this phase the hydrogen peroxide is consumed in an oxidation reaction by a second enzyme, peroxidase, to produce an oxidized phenol dye, Quinonimine:



Note the stoichiometric ratios of each of the components throughout this process.

Other things of note:

Blood glucose concentrations in a healthy person range from: 3.9-5.5mM (70-100 mg/dL in biochemistry)

Oxygen is also a key component in the first reaction. The saturation concentration of dissolved oxygen in water at room temperature is: 0.27 mM

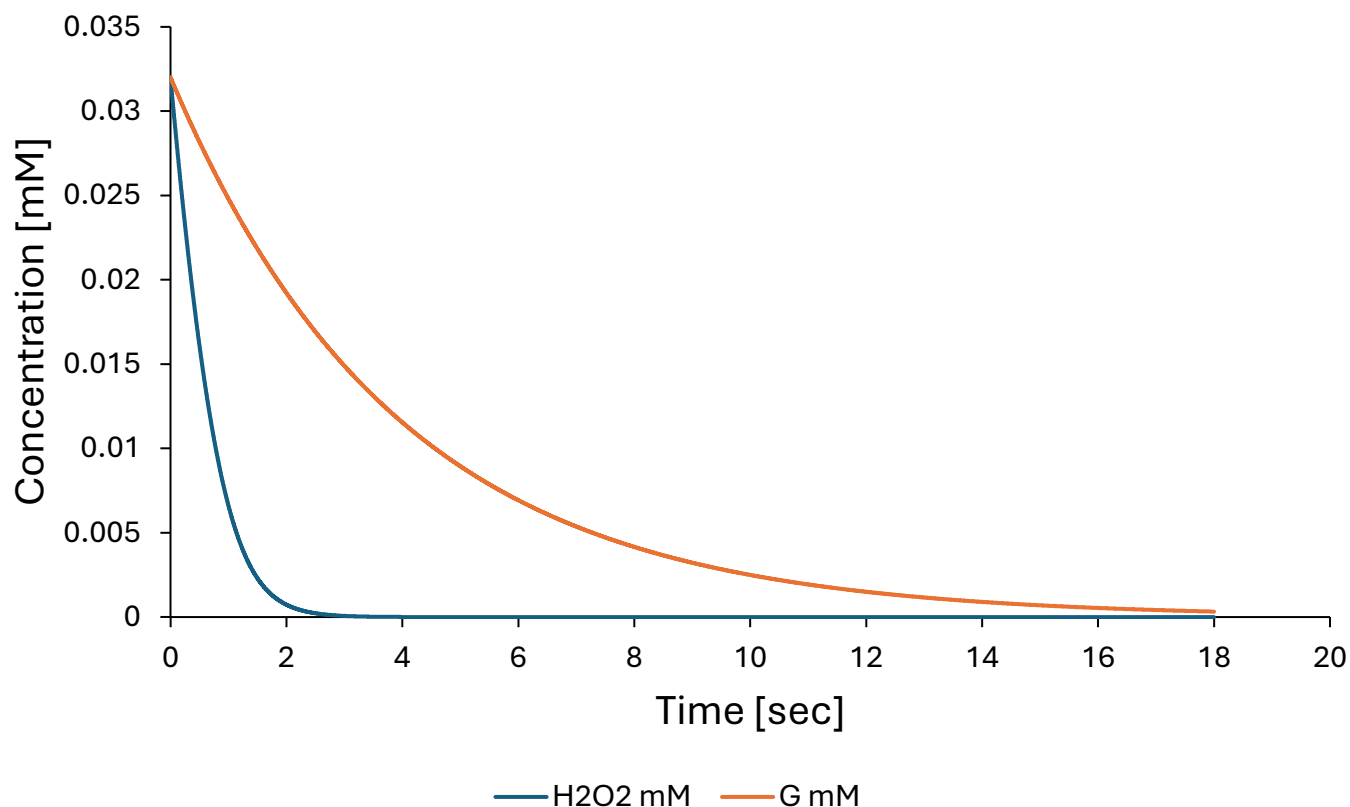
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Probe Reaction:

The second phase of the reaction above is what is called a “probe” reaction. It is an additional reaction in a process like the Trinder reaction, intended to detect the concentration of a solute that is normally colorless or undetectable through normal means. The probe reaction is meant to provide a signal which is readable through the tools available, in this case producing a visible red product which is stoichiometrically proportional to the glucose present in the sample.

How quickly should the probe reaction be relative to the main reaction if we want to learn about said reactions kinetics?

Endpoint Reaction, Glucose Oxidase vs Peroxidase



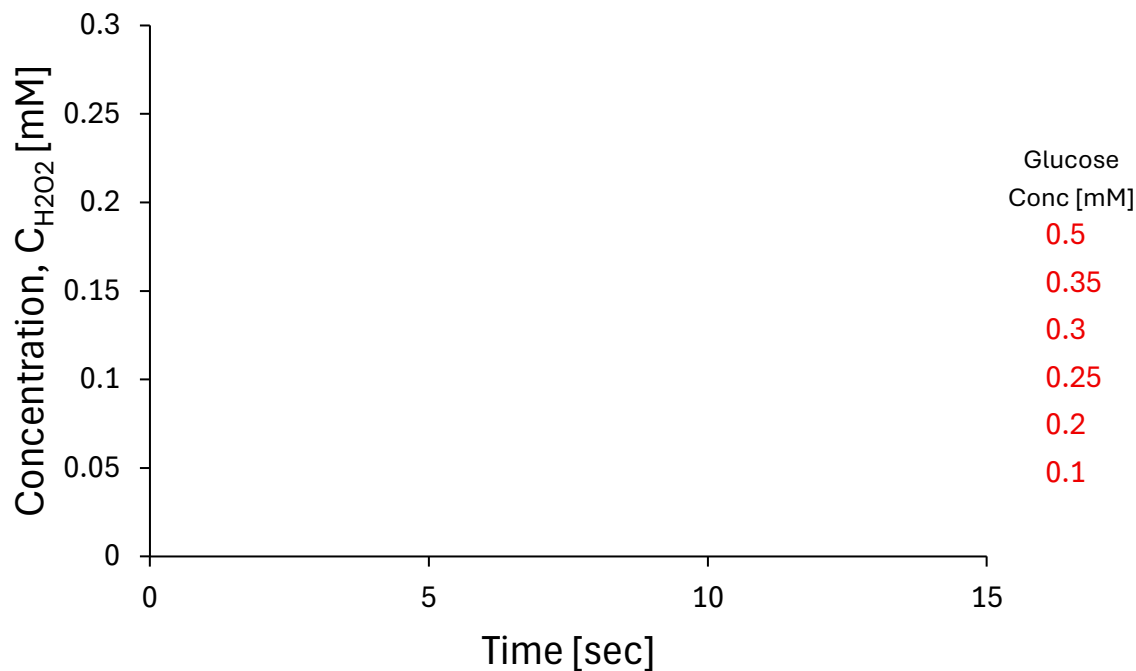
As you can see on this graph, glucose and hydrogen peroxide are made to run in parallel reactions starting at the same initial concentration with their respective enzymes. Notice how the peroxide reaction (the probe reaction) has a much steeper slope than the glucose oxidase reaction.

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Calibration Curves for Glucose Concentrations:

- 1) Given the initial glucose concentrations listed on the graph (in red) and that oxygen (O_2) reaches saturation in water at 0.27 mM at room temperature, roughly plot sketch the graph for peroxide concentration vs time below assuming a start at the origin, (0,0).

Glucose Endpoint Graph (Model)



In the previous activity, we explored the probe reaction:



$$r_{Dye} = \frac{dC_{Dye}}{dt} = -\frac{dC_{H_2O_2}}{dt} = \frac{r_{max}C_{H_2O_2}}{K_M + C_{H_2O_2}}$$

This process consumes the hydrogen peroxide produced by the glucose oxidase reaction examined in this study.

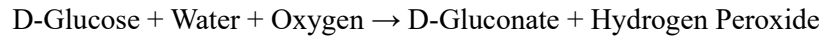
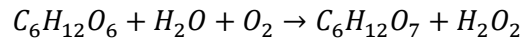
Knowing the reaction from part 1 is referred to as the probe reaction, which of the two reactions is the fastest? Explain your answer.

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Overview of Experiment:

The following experiment is a colorimetric reaction intended to measure the concentration of glucose in an aqueous solution.

Glucose Oxidase Reaction:



$$\frac{dC_G}{dt} = r_g = \frac{k_4 C_E C_G}{\frac{k_4}{k_1} + C_G \left(\frac{k_4}{k_3 C_{O_2}} + \frac{k_4}{k_2 + k_5 C_G} + 1 \right)}$$

k_1, k_2, k_3, k_4, k_5 = Rate Constants,

C_G = Glucose Concentration,

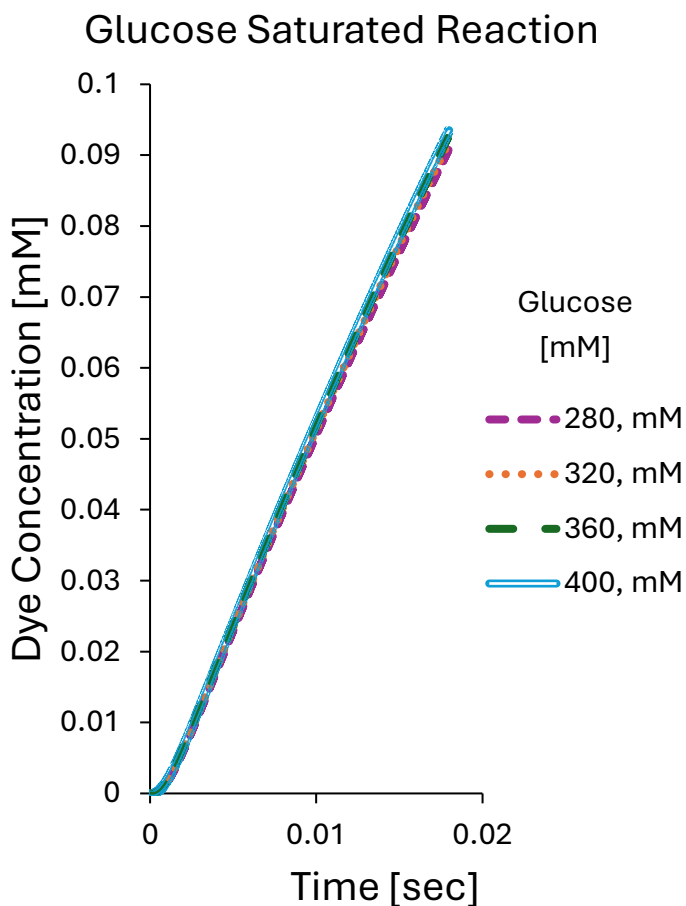
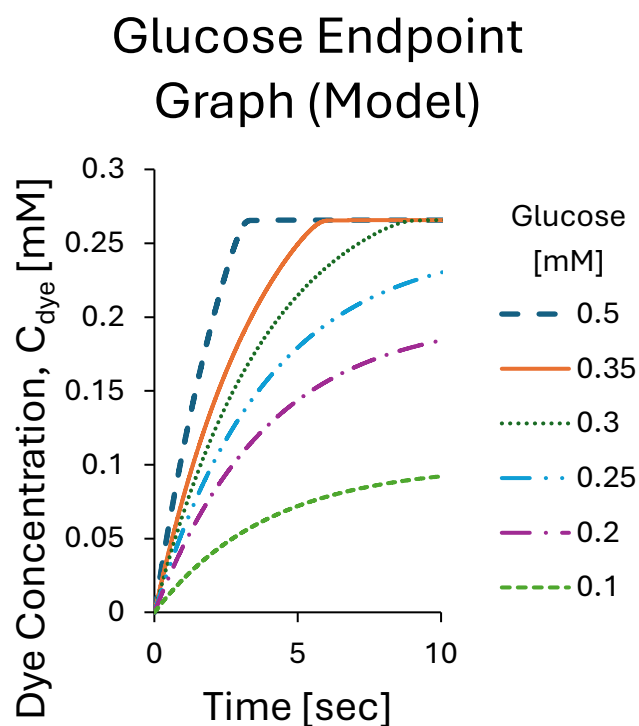
C_E = Enzyme Concentration,

C_{O_2} = Oxygen Concentration

The hydrogen peroxide is then consumed during the second reaction (see Part I activity).

These two processes together make up what is called the Trinder reaction.

Below is a graph with simulated data, showing C_{Dye} [mM] vs. t [sec] for different concentrations of glucose, assuming O_2 is limited to the solubility concentration of 0.27 mM:



Discuss trends in initial rates depicted in the graph.

How are the endpoints and slopes of each graph affected by initial concentration?

What does this say about the law of mass action?

What do you learn about enzyme catalyst turnover rate, i.e., r_{max} ?

Constructing a Calibration Curve:

Step 1) Follow the set-up video for your kit. Wear gloves. Clean up spills.

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Step 2) Prime your cellphone camera. One group member should oversee this.

Step 3) Use gloves and safety glasses for handling chemicals; record sample glucose concentration. One group member should oversee chemical handling, one should record values.

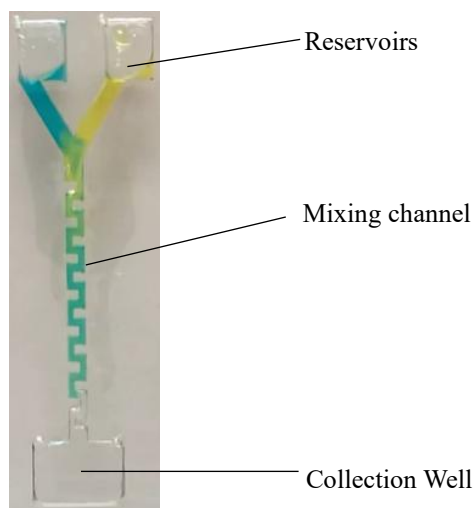
Step 4) When ready, load the glucose sample and enzyme solution into separate reservoirs to start each trial.

- Note your solutions are colorless; we used dyed water to show there are two separate liquids loaded into separate reservoirs.
- To prevent premature reaction from leftover solutions, always load sample solution on top, reagent solution on bottom for repeats.



Step 5) Turn on the flashlight behind the cardboard box & insert your phone.

Step 6) Start a timer on a cell phone exactly when a teammate places module in an upright position with reservoirs at the top.



- If needed shake the module to allow the solutions to flow to the bottom

Step 7) After loading, record the G-value from the RGB readout. Place at time $t = 0$ seconds.

Step 8) Continue recording the G-value every thirty seconds for three minutes.

Step 9) Repeat steps 6-8 for a second trial. Average your results.

Step 10) Record measurements for a blank by collecting the signal from just the enzyme solution and buffer.

Step 11) Share results with your professor so all sample results can be shown to the rest of the class. Make sure to label their concentration when filling in the results (mM)

Measure the signal from sample three times and record here:

Time [Sec]	Blank	Sample ____ Trial 1 (Yours)	Sample ____ Trial 2 (Yours)	Average	Sample 2	Sample 3	Sample 4	Sample 5
0								

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30								
60								
90								
120								
150								
180								

The G-value (G) correlates to the intensity (I) of green light allowed to pass through the red sample. Knowing the Probe reaction rate $r_{Dye} = \left. \frac{dC_{Dye}}{dt} \right|_{t=0} = \frac{r_{max}C_{H_2O_2}}{K_M + C_{H_2O_2}}$ is $\gg \left. \frac{-dC_{Glucose}}{dt} \right|_{t=0}$, as we learned above. So, we can assume a 1:1 instantaneous conversion of any H_2O_2 formed to red 2nd Probe reaction:

$$xG = I$$

$$ABS = -\log T = -\log\left(\frac{I}{I_0}\right) = -\log\left(\frac{xG}{xG_0}\right)$$

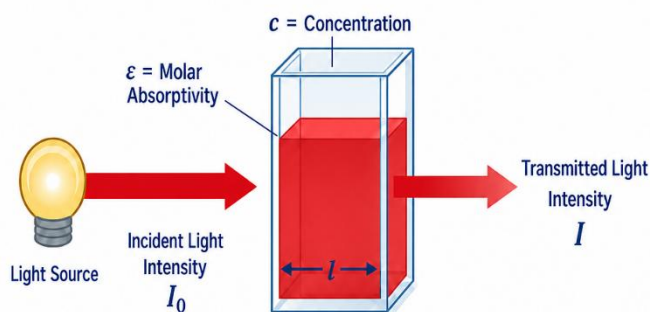
G_0 = Blank G-value

I_0 = Blank Intensity

ABS = absorbance

T = transmittance

x = Correction factor for camera (assumed constant)



$$A = \epsilon l c = -\log_{10} \frac{I}{I_0}$$

Beer-Lambert Law

A = Absorbance (unitless)

ϵ = Molar absorptivity ($L \cdot mol^{-1} \cdot cm^{-1}$)

l = Path length (cm)

c = Concentration ($mol \cdot L^{-1}$)

I_0 = Incident light intensity

I = Transmitted light intensity

Beer's Law:

$$ABS = \epsilon \cdot L \cdot C = -\log(T)$$

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C = concentration (known)

L = pathlength (0.63 cm)

ϵ = molar absorptivity ($\sim 19.6 \text{ mM}^{-1}\text{cm}^{-1}$)

Record Observations

Earlier we assumed the G-value (G) and the light intensity (I) were proportional. How likely is this to be true? Explain.

When constructing a calibration curve, choosing a blank to compare a sample to is imperative. Why? Briefly explain.

Homework Problems:

- 1) Convert absorbance to dye concentration using Beer's Law to fill in the following table:

Time [sec]	Sample 1 [mM]	Sample 2	Sample 3	Sample 4	Sample 5
0					
30					
60					
90					
120					
150					
180					

Do these values match up with what you expect (think initial concentrations)?

- 2) Roughly estimate the initial rate of reaction for each concentration using the first-time interval of 30 sec.

Does the reaction approach a maximum rate of reaction, like with Part 1? Explain.

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Up to a point, increasing the concentration of reactants, such as glucose, should hasten the reaction rate.

The reaction rate, or the rate of consumption of glucose as derived by Duke et al, is shown below.

$$\frac{dC_G}{dt} = r_g = \frac{k_4 C_E C_G}{\frac{k_4}{k_1} + C_G \left(\frac{k_4}{k_3 C_{O_2}} + \frac{k_4}{k_2 + k_5 C_G} + 1 \right)}$$

It's worth noting that the relationship between k_1 and the reaction rate is:

$$r_1 = k_1 C_E C_G$$

$$r_g = -r_1$$

Knowing this,

- 3) Plot the reaction rate versus the initial concentration of glucose and find k_1 given $C_E = 3184 \text{ M}$. Write your answer below.

- 4) Using the data provided below, please determine the other rate constants. You should have k_1 from problem 3. Next find k_2 , k_3 , k_4 and k_5 by fitting data at an intermediate C_G and using an Excel solver. Compare to the literature values given on completion (Note: Perfect accuracy is not expected).

Given Data:

Glucose, mM = 0.1

Oxygen, mM = 0.27

Enzyme, mM = 0.02

Rate, $\frac{dG}{dt} = -0.025$

Input the given data in one column. These are constants.

In a separate column, put your answer for k_1 and input a guess for k_2 , k_3 , k_4 and k_5 . These are your variables.

Finally, in a third column, enter the full rate equation shown above with your variables and constants as inputs. It should look something like this:

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AutoSave On Solver Experiment- Give it a Go! • Saved

File Home Developer Insert Draw Page Layout Formulas Data Review View Automate

Clipboard Font Alignment Number Styles Cells Editing

Formula Bar: $=-(\$L\$4*\$I\$2*\$I3)/((\$L\$4/\$L\$1)+I3*((\$L\$4/(\$L\$3*\$I\$1))+(\$L4/(\$L\$2+\$L\$5*\$I3))+1))$

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1								Oxygen	0.265625	mol/m3	k1	10	m3/mol/s	rg	1))			
2								Enzyme	0.021875	mol/m3	k2	11785649	1/s	err	1.33E-05			
3								Glucose	0.1	mol/m3	k3	53687092	m3/mol/s					
4											k4	14260635	1/s					
5								dG/dt	-0.02552	mM/s	k5	1178566	m3/mol/s					
6																		
7																		
8																		
9																		
10																		
11																		

Include a cell for the error, $(\text{measured rate} - \text{predicted rate})^2$. This is what solver will attempt to minimize by varying the rate constants. Make sure to constrain the system by stating your variables must be positive.